6 Average

IMPORTANT FACTS AND FORMULAE

I. Average $= \left(\dfrac{\text{Sum of observations}}{\text{Number of observations}}\right)$

II. Suppose a man covers a certain distance at x kmph and an equal distance at y kmph. Then, the average speed during the whole journey is $\left(\dfrac{2xy}{x+y}\right)$ kmph. office 365 off

SOLVED EXAMPLES

Ex. 1. ***Find the average of the following set of numbers: 354, 281, 623, 518, 447, 702, 876.*** **(Bank Recruitment, 2009)**

Sol. Average of given numbers $= \left(\dfrac{354+281+623+518+447+702+876}{7}\right) = \dfrac{3801}{7} = 543.$

Ex. 2. ***The body weight of six boys is recorded as 54 kg, 64 kg, 75 kg, 67 kg, 45 kg and 91 kg. What is the average body weight of all six boys?*** **(Bank Recruitment, 2010)**

Sol. Average body weight $= \left(\dfrac{54+64+75+67+45+91}{6}\right)$ kg $= \left(\dfrac{396}{6}\right)$ kg $= 66$ kg.

Ex. 3. ***There are six numbers 30, 72, 53, 68, x and 87, out of which x is unknown. The average of the numbers is 60. What is the value of x?*** **(Bank Recruitment, 2010)**

Sol. Average of given numbers $= \left(\dfrac{30+72+53+68+x+87}{6}\right) = \left(\dfrac{310+x}{6}\right).$

$\therefore \quad \dfrac{310+x}{6} = 60 \Rightarrow 310 + x = 360 \Rightarrow x = 50.$

Hence, $x = 50$.

Ex. 4. ***Find the average of all prime numbers between 30 and 50.***

Sol. There are five prime numbers between 30 and 50.

They are 31, 37, 41, 43 and 47.

$\therefore$ Required average $= \left(\dfrac{31+37+41+43+47}{5}\right) = \dfrac{199}{5} = 39.8.$

Ex. 5. ***Find the average of first 40 natural numbers.***

Sol. Sum of first n natural numbers $= \dfrac{n(n+1)}{2}.$

So, sum of first 40 natural numbers $= \dfrac{40 \times 41}{2} = 820.$

$\therefore$ Required average $= \dfrac{820}{40} = 20.5.$

Ex. 6. ***Find the average of first 20 multiples of 7.***

Sol. Required average $= \dfrac{7(1+2+3+.....+20)}{20} = \left(\dfrac{7 \times 20 \times 21}{20 \times 2}\right) = \left(\dfrac{147}{2}\right) = 73.5.$

Ex. 7. ***A man bought 5 shirts at ₹ 450 each, 4 trousers at ₹ 750 each and 12 pairs of shoes at ₹ 750 each. What is the average expenditure per article?*** **(R.R.B., 2006)**

Sol. Total expenditure = ₹ (5 × 450 + 4 × 750 + 12 × 750) = ₹ (2250 + 3000 + 9000) = ₹ 14250.